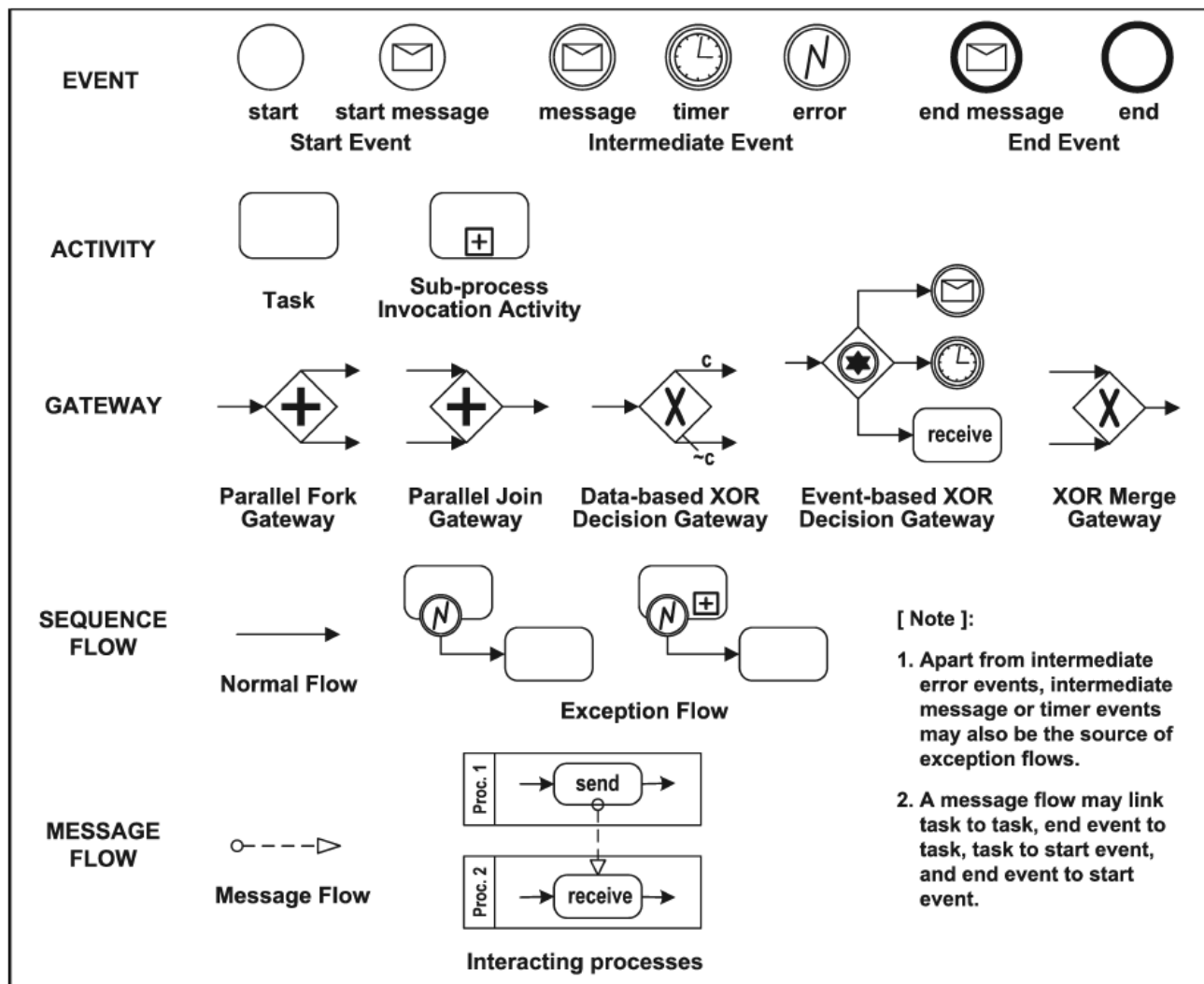


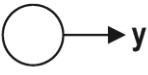
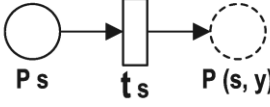
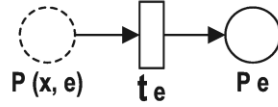
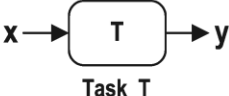
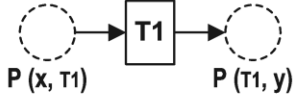
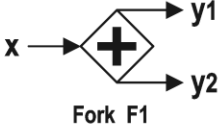
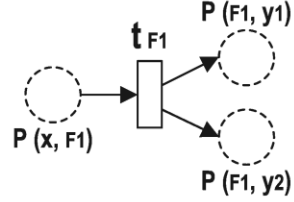
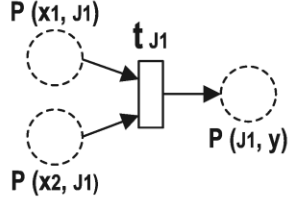
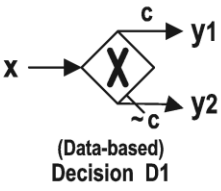
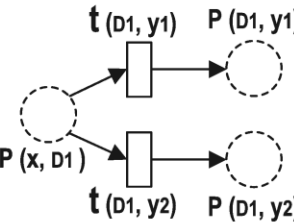
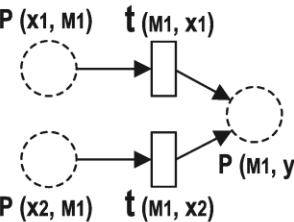
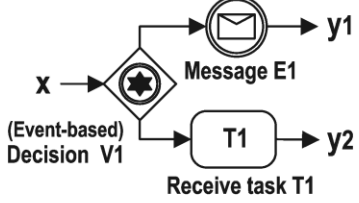
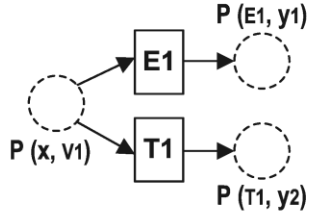
BPMN to Petri nets Mapping (Introduction)

- Core subset of BPMN elements

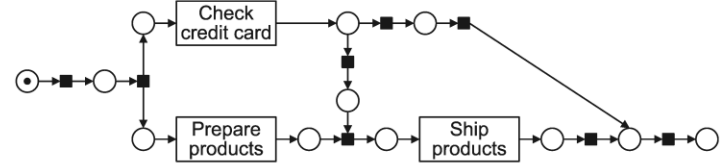
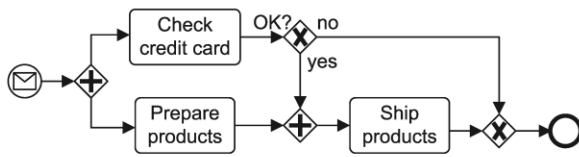


- Note that** there are different representations of exception flow based on the type of interruption provided in the intermediate event. If it is interrupting event, then its logic is similar to “data-based XOR” gateway. Otherwise, non-interrupting, then its logic is similar to “parallel fork” gateway.

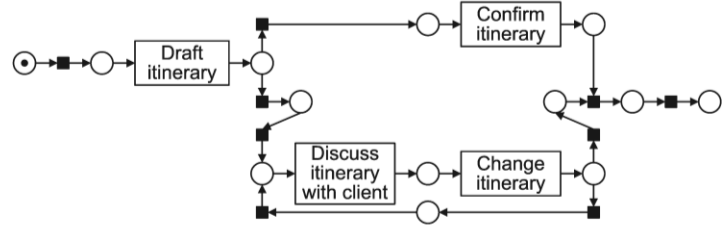
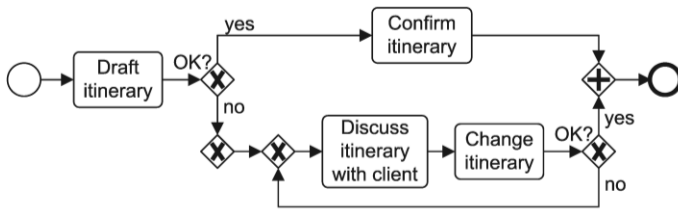
Mapping task, events, and gateways onto Petri-net modules.

BPMN Object	Petri-net Module	BPMN Object	Petri-net Module
 Start s	 P_s t_s $P(s, y)$	 End e	 $P(x, e)$ t_e P_e
 Message E	 $P(x, E1)$ $E1$ $P(E1, y)$	 Task T	 $P(x, T1)$ $T1$ $P(T1, y)$
 Fork F1	 $P(x, F1)$ t_{F1} $P(F1, y1)$ $P(F1, y2)$	 Join J1	 $P(x1, J1)$ $P(x2, J1)$ t_{J1} $P(J1, y)$
 (Data-based) Decision D1	 $P(x, D1)$ $t(D1, y1)$ $t(D1, y2)$ $P(D1, y1)$ $P(D1, y2)$	 Merge M1	 $P(x1, M1)$ $P(x2, M1)$ $t(M1, x1)$ $P(M1, y)$
 (Event-based) Decision V1 Message E1 Receive task T1	 $P(x, V1)$ $E1$ $T1$ $P(E1, y1)$ $P(T1, y2)$	[Note]: x, x1 or x2 represents an input object, and y, y1 or y2 represents an output object.	

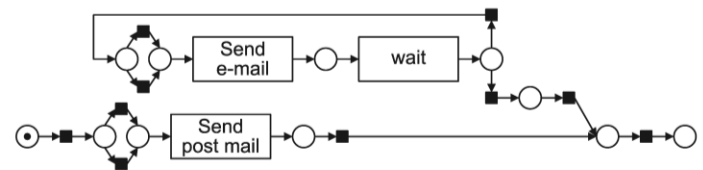
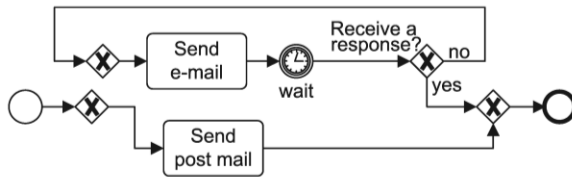
- Examples of BPMN process models and transformations to Petri nets.



(a) Order process

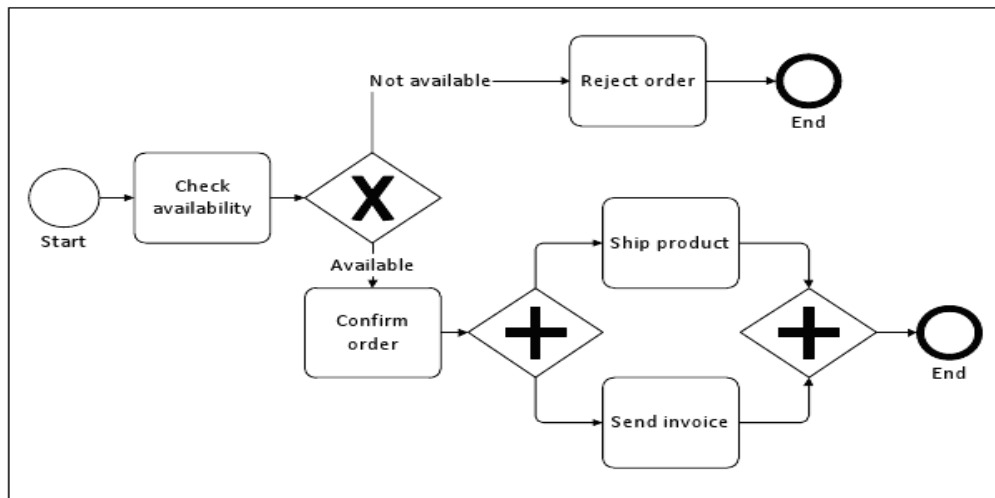


(b) Travel itinerary process

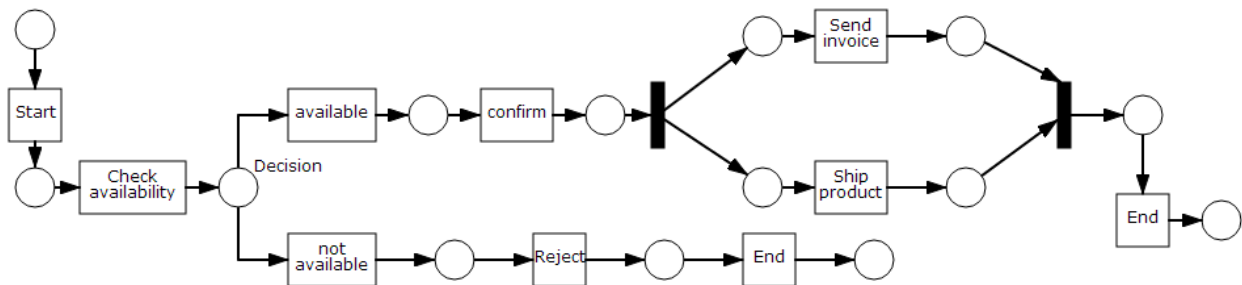
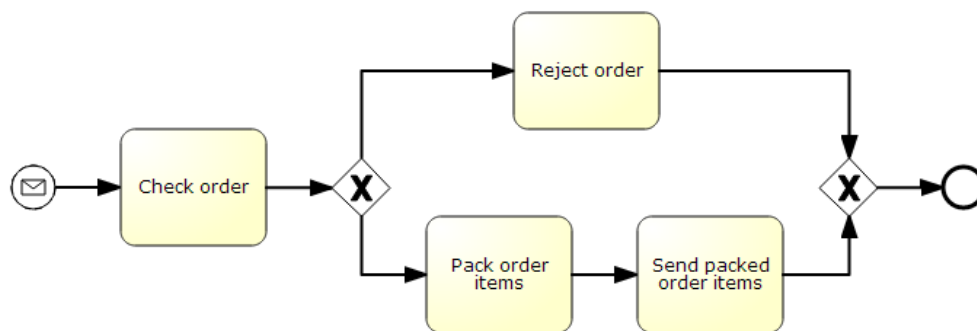


(c) Answer process

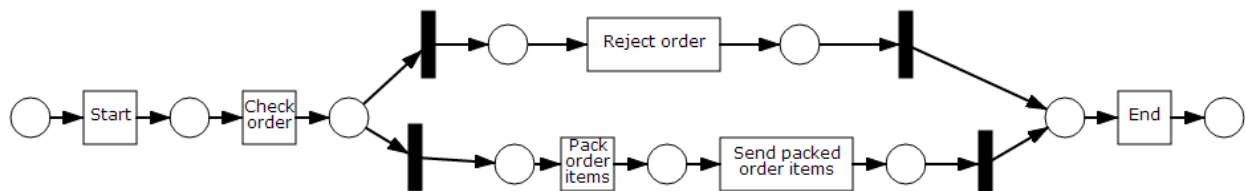
Note: ■ silent transition

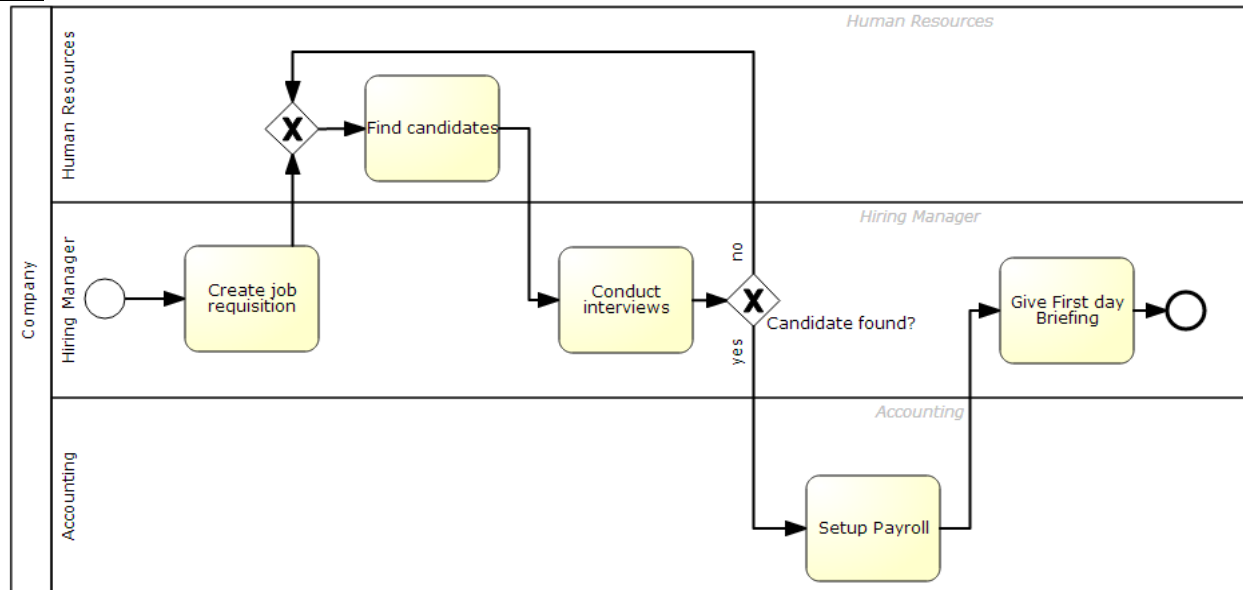
Practice Examples:**Example 1:**

Transformation to Petri nets:

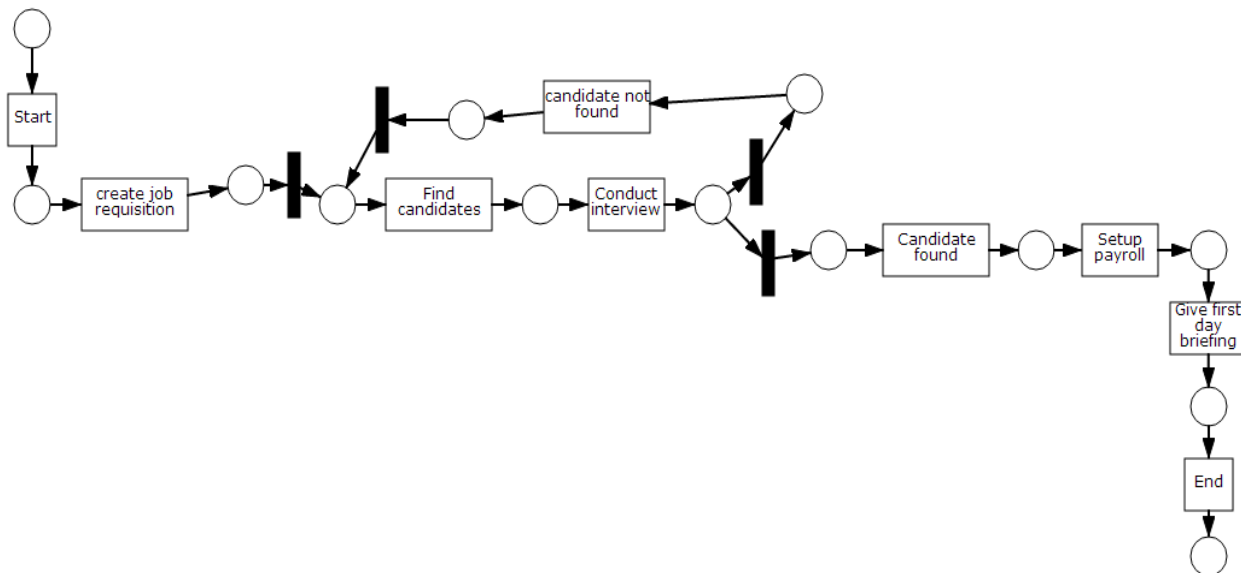
**Example 2:**

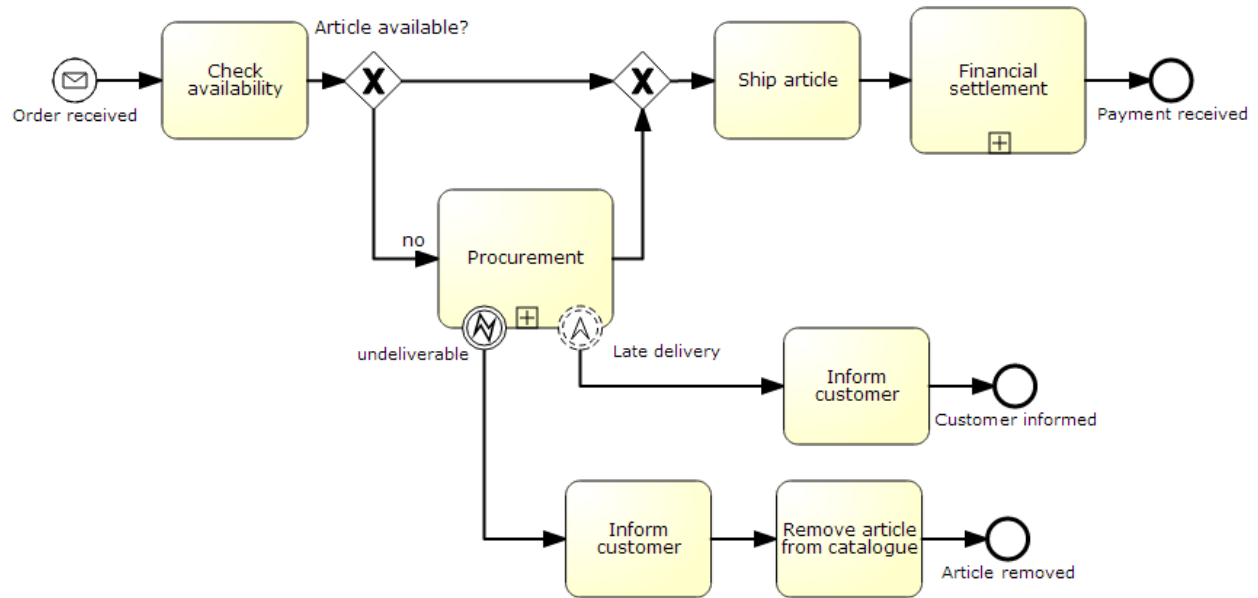
Transformation to Petri nets:



Example 3:

Transformation to Petri nets:



Example 4:

Transformation to Petri nets:

